VitaminC

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```
library(car)
```

```
## Loading required package: carData
```

```
library(HH)
```

```
## Loading required package: lattice
## Loading required package: grid
## Loading required package: latticeExtra
## Loading required package: RColorBrewer
## Loading required package: multcomp
## Loading required package: mvtnorm
## Loading required package: survival
## Loading required package: TH.data
## Loading required package: MASS
##
## Attaching package: 'TH.data'
## The following object is masked from 'package:MASS':
##
##     geyser
## Loading required package: gridExtra
##
## Attaching package: 'HH'
## The following objects are masked from 'package:car':
##
##     logit, vif
```

```
library(car)
library(emmeans)
```

```
##
## Attaching package: 'emmeans'
## The following object is masked from 'package:HH':
##
##     as.glht
## The following object is masked from 'package:multcomp':
##
##     cld
```

```
library(tables)
```

```
## Loading required package: Hmisc
```

```
## Loading required package: Formula
## Loading required package: ggplot2
##
## Attaching package: 'ggplot2'
## The following object is masked from 'package:latticeExtra':
##
##     layer
##
## Attaching package: 'Hmisc'
## The following objects are masked from 'package:base':
##
##     format.pval, units
```

```
library(RcmdrMisc)
```

```
## Loading required package: sandwich
##
## Attaching package: 'RcmdrMisc'
## The following object is masked from 'package:Hmisc':
##
##     Dotplot
```

ANCOVA

We want to COMPARE TWO PEDAGOGICAL METHODOLOGIES.

Loading the data and printing the top part

```
setwd("G:/PiE2/2018")
vitamincdata<-read.csv2("./Dades/vitc.csv")
head(vitamincdata)
```

```
##   treat week vitc
## 1     a    1 30.2
## 2     a    2 29.2
## 3     a    3 23.8
## 4     a    4 27.4
## 5     a    5 16.8
## 6     a    6 29.6
```

```
dim(vitamincdata)
```

```
## [1] 72  3
```

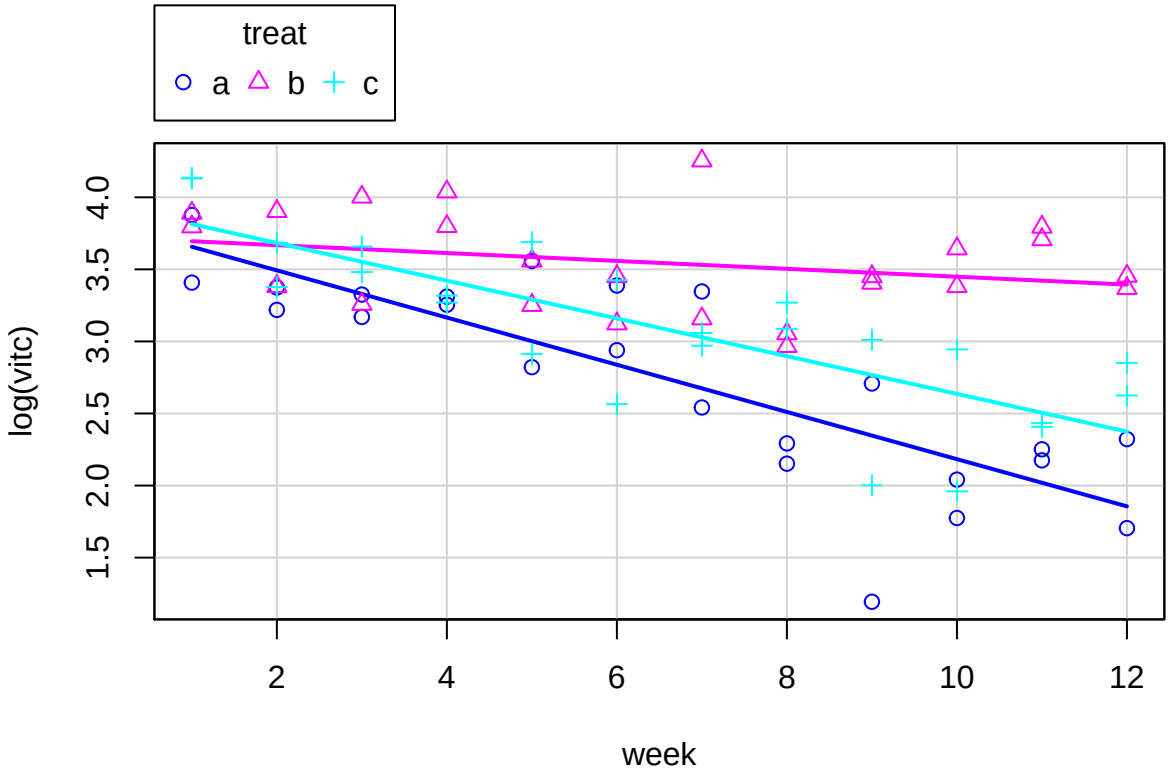
The dataset has 72 rows and three columns. Each row corresponds to a fix orange juice. The columns specify: the treatment or conservation method which is a categorical variable with three levels: a, b, and c, the week which is also a categorical variable with 12 levels, indicating the week in which the measure has been taken and finally, the vitaminC level of the orange juice.

The response of dependent variable is the vitaminC level and the covariates or explanatory variables are the conservation method and the week.

The most important question is to known if the orange juices lose the vitaminC in a similar way for the different consevation methods.

Descriptive statistics

```
scatterplot(log(vitc)~week|treat,smooth=F,dat=vitamincdata)
```



The model

```
mvitaminc<-lm(log(vitc)~week+treat:week,vitamincdata)

summary(mvitaminc)
```

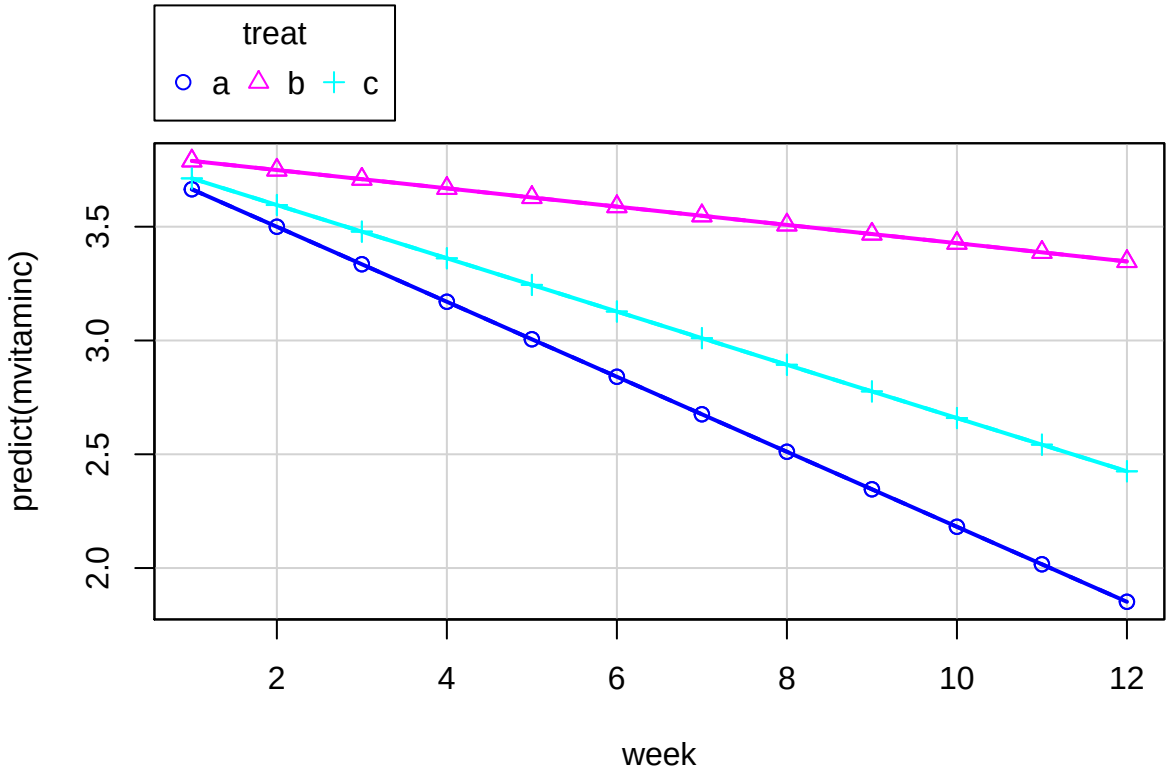
```
##
## Call:
## lm(formula = log(vitc) ~ week + treat:week, data = vitamincdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.15221 -0.19240  0.00464  0.23452  0.70470
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.82934    0.09048  42.324  < 2e-16 ***
```

```
## week        -0.16480    0.01475 -11.171  < 2e-16 ***
## week:treatb  0.12462    0.01412   8.823 7.04e-13 ***
## week:treatc  0.04778    0.01412   3.383  0.00119 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3601 on 68 degrees of freedom
## Multiple R-squared:  0.6957, Adjusted R-squared:  0.6823
## F-statistic: 51.83 on 3 and 68 DF,  p-value: < 2.2e-16
```

```
Anova(mvitaminc,ty=3)
```

```
## Anova Table (Type III tests)
##
## Response: log(vitc)
##              Sum Sq Df F value    Pr(>F)
## (Intercept) 232.275  1 1791.33 < 2.2e-16 ***
## week         16.182  1  124.80 < 2.2e-16 ***
## week:treat   10.277  2   39.63 3.894e-12 ***
## Residuals     8.817 68
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
scatterplot(predict(mvitaminc)~week|treat,dat=vitamincdata)
```



Le us compare the slopes

```
(emmm<-emtrends(mvitaminc,~treat,var="week"))
```

```
##  treat  week.trend       SE df    lower.CL    upper.CL
##  a     -0.16480078 0.014752 68 -0.19423795 -0.13536361
##  b     -0.04018146 0.014752 68 -0.06961863 -0.01074430
##  c     -0.11701781 0.014752 68 -0.14645497 -0.08758064
##
## Trends are based on the log (transformed) scale
## Confidence level used: 0.95
```

```
pairs(emmm)
```

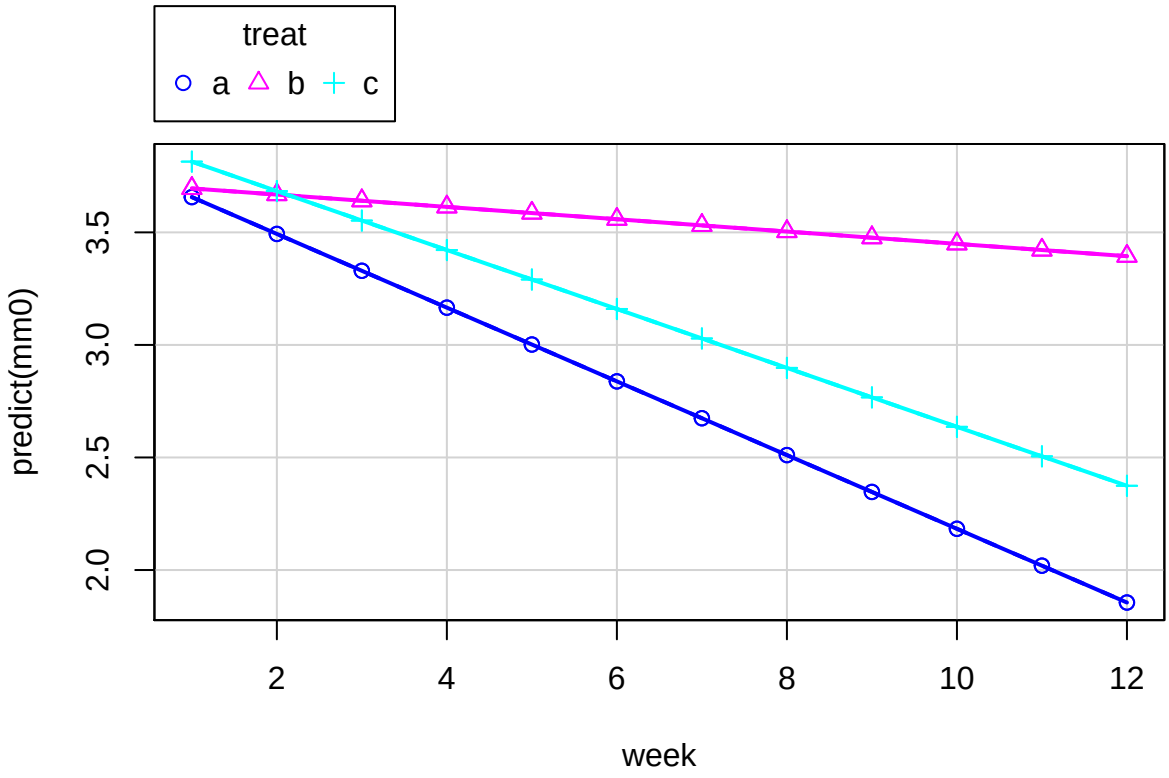
```
##  contrast    estimate         SE df t.ratio p.value
##  a - b    -0.12461932 0.01412397 68  -8.823  <.0001
##  a - c    -0.04778297 0.01412397 68  -3.383  0.0034
##  b - c     0.07683634 0.01412397 68   5.440  <.0001
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

Model with straight lines that may be different at week zero

```
summary(mm0<-lm(log(vitc)~week*treat,vitamincdata))
```

```
##
## Call:
## lm(formula = log(vitc) ~ week * treat, data = vitamincdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.15293 -0.18979 -0.01522  0.24540  0.72179
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.82038    0.15788  24.199  < 2e-16 ***
## week        -0.16373    0.02145  -7.632 1.20e-10 ***
## treatb      -0.09785    0.22327  -0.438    0.663
## treatc       0.12472    0.22327   0.559    0.578
## week:treatb  0.13636    0.03034   4.495 2.88e-05 ***
## week:treatc  0.03282    0.03034   1.082    0.283
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3628 on 66 degrees of freedom
## Multiple R-squared:  0.7003, Adjusted R-squared:  0.6776
## F-statistic: 30.84 on 5 and 66 DF,  p-value: 4.858e-16
```

```
scatterplot(predict(mm0)~week|treat,dat=vitamincdata)
```




```
Anova(mm0,ty=3)
```

```
## Anova Table (Type III tests)
##
## Response: log(vitc)
##             Sum Sq Df  F value    Pr(>F)
## (Intercept) 77.063  1 585.5676 < 2.2e-16 ***
## week         7.667  1  58.2546 1.204e-10 ***
## treat        0.131  2   0.4992    0.6093
## week:treat   2.897  2  11.0081 7.488e-05 ***
## Residuals    8.686 66
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Multiple comparison fo the three terms

```
(emmt<-emmeans(mm0,~treat|week,at=list(week=c(0))))
```

```
## week = 0:
##  treat   emmean        SE df lower.CL upper.CL
##  a     3.820383 0.1578768 66 3.505172 4.135595
##  b     3.722537 0.1578768 66 3.407326 4.037748
##  c     3.945101 0.1578768 66 3.629890 4.260312
##
## Results are given on the log (not the response) scale.
## Confidence level used: 0.95
```

```
print(pairs(emmt))
```

```
## week = 0:
##  contrast    estimate        SE df t.ratio p.value
##  a - b     0.09784614 0.2232716 66   0.438  0.8997
##  a - c    -0.12471756 0.2232716 66  -0.559  0.8424
##  b - c    -0.22256370 0.2232716 66  -0.997  0.5815
##
## Results are given on the log (not the response) scale.
## P value adjustment: tukey method for comparing a family of 3 estimates
```

Multiple comparison of the three slopes allowing different intercepts

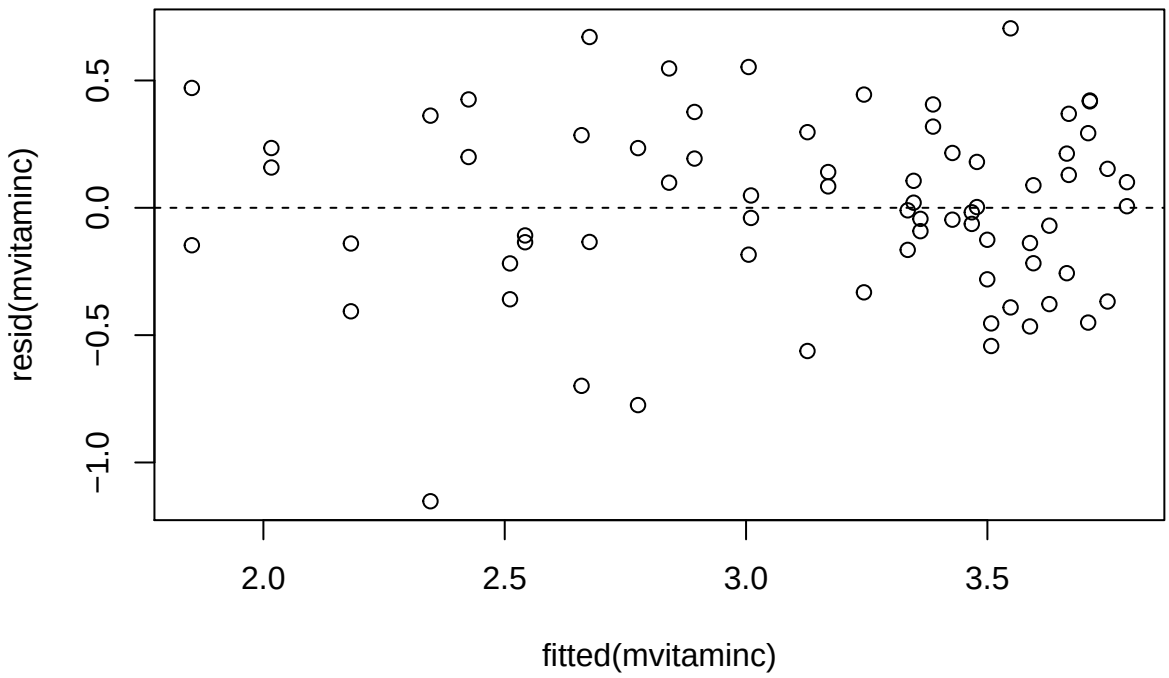
```
(emmm<-emtrends(mm0,~treat,var="week"))
```

```
##  treat  week.trend         SE df    lower.CL    upper.CL
##  a     -0.16372592 0.02145124 66 -0.20655470 -0.12089714
##  b     -0.02736507 0.02145124 66 -0.07019385  0.01546371
##  c     -0.13090906 0.02145124 66 -0.17373783 -0.08808028
##
## Trends are based on the log (transformed) scale
## Confidence level used: 0.95
```

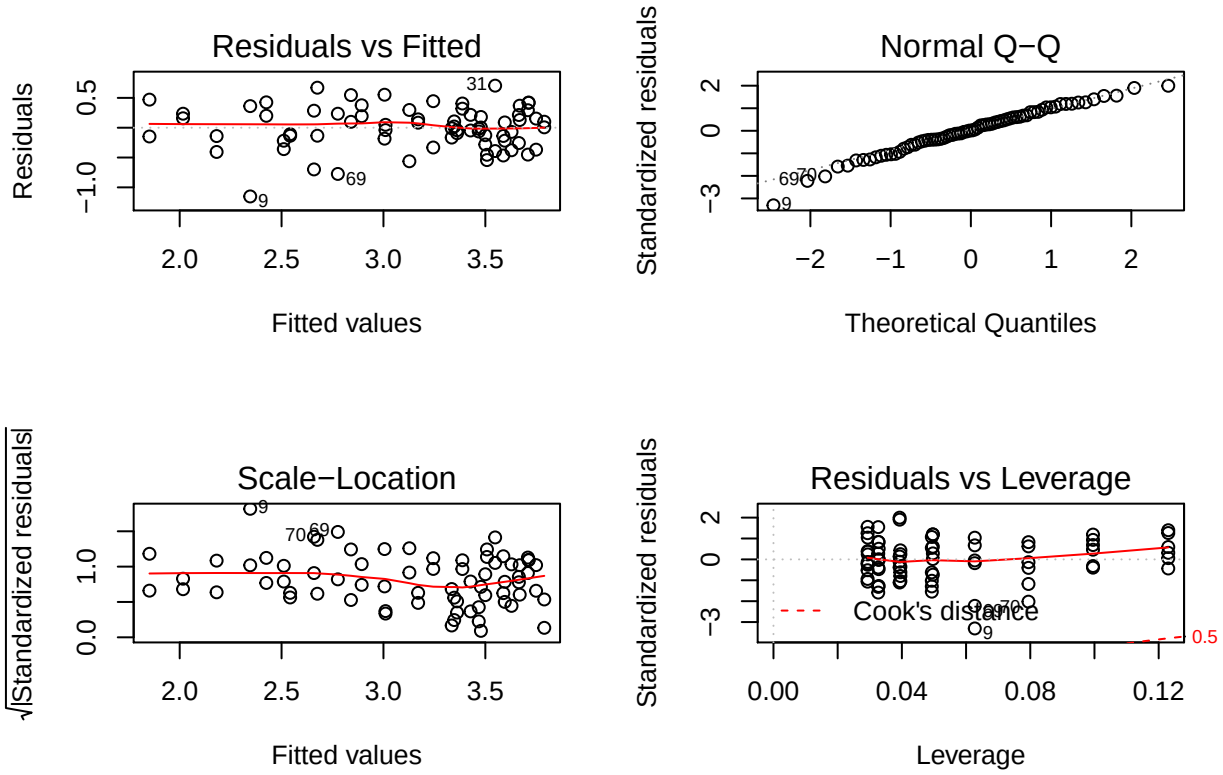
```
print(pairs(emmm))
```

```
##  contrast    estimate         SE df t.ratio p.value
##  a - b    -0.13636085 0.03033663 66  -4.495  0.0001
##  a - c    -0.03281687 0.03033663 66  -1.082  0.5287
##  b - c     0.10354399 0.03033663 66   3.413  0.0031
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
plot(fitted(mvitaminc),resid(mvitaminc))
abline(h=0,lty=2)
```

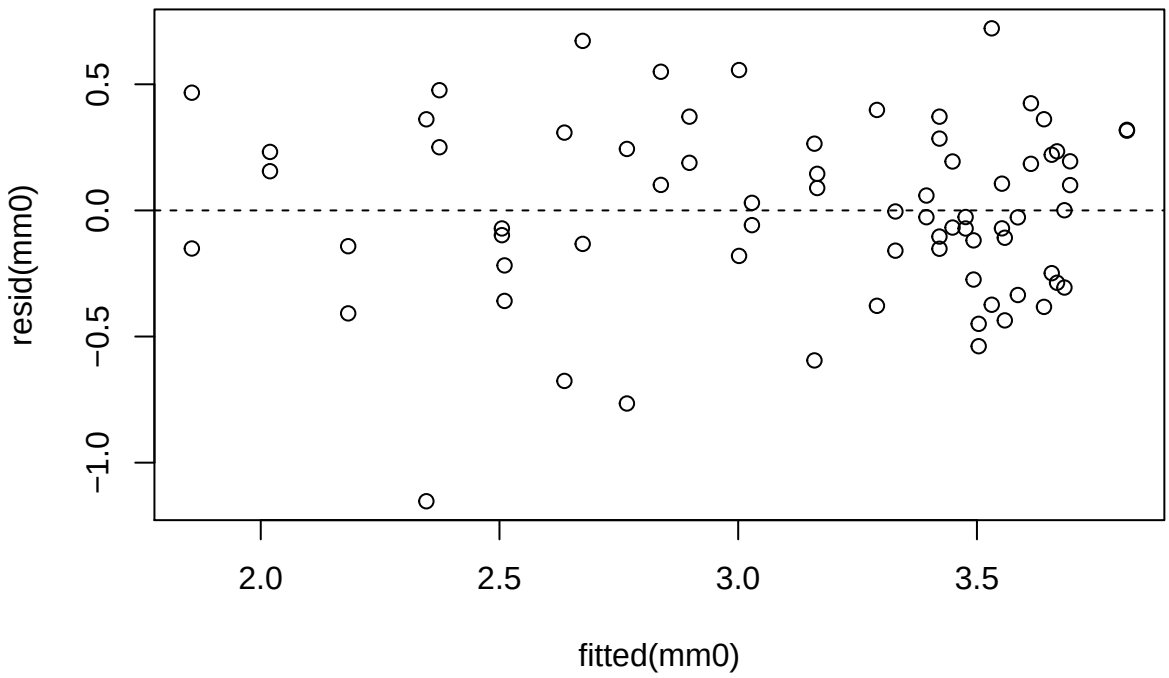



```
oldpar<-par(mfrow=c(2,2))
plot(mvitaminc,ask=F)
```

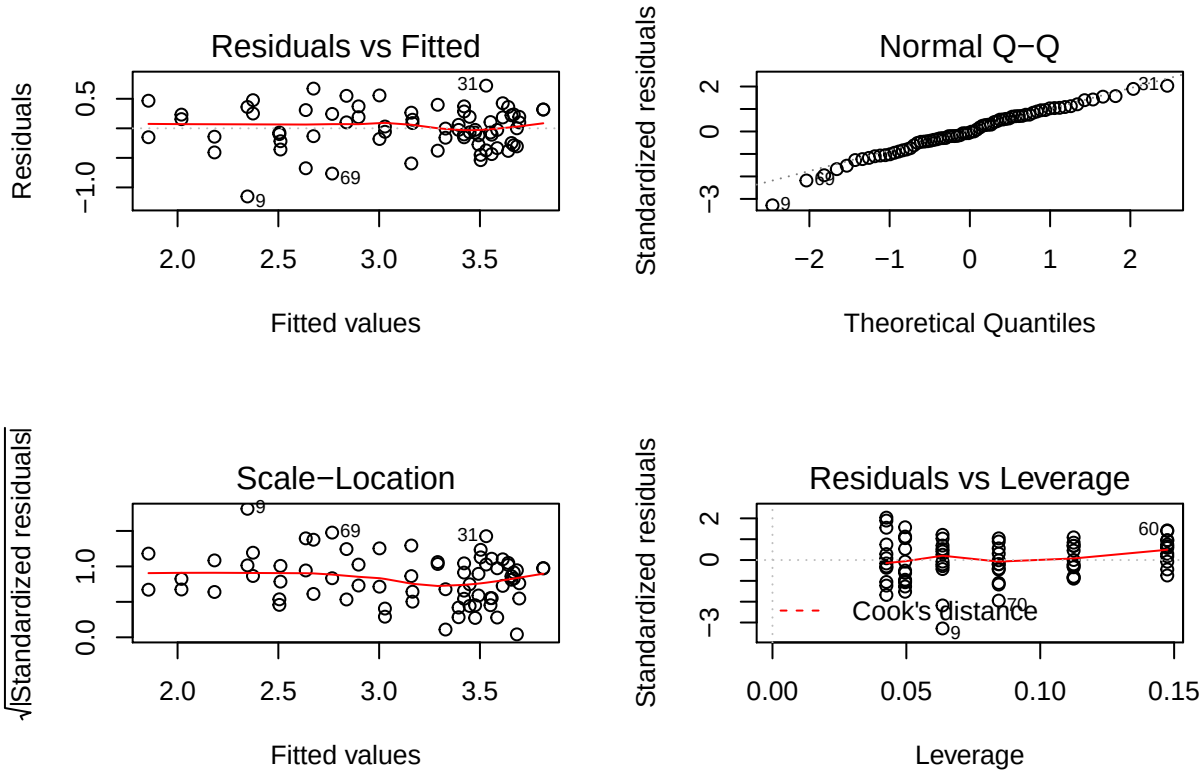



```
par(oldpar)

plot(fitted(mm0),resid(mm0))
abline(h=0,lty=2)
```




```
oldpar<-par(mfrow=c(2,2))
plot(mm0,ask=F)
```


Residuals vs Fitted
Residuals
Fitted values
Normal Q−Q
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage

```
par(oldpar)
```